4, 1939, the day after Britain had declared war on Germany. He worked in a small team of mathematicians and code-breakers who had already enjoyed some success in deciphering Enigma messages.

Using logic and probability, Turing realized that to find out all the possible combinations of the Enigma code, he had to build a computer that was capable of replicating 60 Enigma machines. To that end, he and colleague Gordon Welchman increased the power of the "Bombe"—a computer constructed by Polish cryptographers in 1938. The first improved Bombe was ready in August 1940, and shortly after that, the Enigma messages started to be deciphered. By the end of the war, more than 200 Bombes were operating day and night at Bletchley Park, even cracking the more complex U-boat ciphers.

Digital computing

Turing was also on hand at Bletchley to observe the development in 1944 of the "Colossus" code-breaking machines, the first programmable digital electronic computers. He immediately saw the

CRACKED THE **150 QUINTILLION** COMBINATIONS OF ENIGMA

HELPED TO DECIPHER **3,000** GERMAN MILITARY MESSAGES A DAY

Enigma (far left) *was an encryption device used by the German military from the 1920s. By World War II, it had became so sophisticated that the Germans believed that their codes could never be broken. However, the Allies broke them with their own computing machine—the Turing-Welchman Bombe (above).*